

Gouranga Pal

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Kolkata, West Bengal, India – 700162

Technical Skills

- **Core CS:** Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Computer Networks, DBMS
- **Languages:** Python, Java, C++, C, C#, JavaScript, Dart, PHP
- **AI / ML / DL:** Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, LLMs, Prompt Engineering, TensorFlow, TensorFlow Lite, NumPy, Pandas, Scikit-learn, Matplotlib
- **Systems & Web:** Flask, REST APIs, React, Cloud Deployment (Firebase/Supabase), Git, Docker, CI-style testing (pytest)
- **Databases:** MySQL, MongoDB, Firebase, Supabase (PostgreSQL)
- **IoT & Embedded:** ESP32, Arduino, Sensor Integration, Edge AI Deployment

Experience

Junior Machine Learning Engineer

May 2024 – Dec 2024

Insighthub Solution

Remote

- Designed, trained, and evaluated ML/DL models in Python (Colab, Jupyter), improving [prediction accuracy / inference time] by [X%] against the prior baseline.
- Partnered with a cross-functional team using JIRA/Agile sprints, shipping [N] model releases across [N] two-week cycles with zero rollback incidents.

Projects

InterVU – AI Mock Interview Platform

Jan 2026 – May 2026

Python, Flask, LLM, Prompt Engineering, Supabase

GitHub | Live Demo

- Architected an LLM-integrated mock-interview platform generating dynamic questions and real-time feedback, cutting manual interview-prep review time by [X%].
- Designed secure auth, scheduling, and analytics with Flask + Supabase (PostgreSQL), scaling to [N]+ concurrent user sessions.

CropCure – Rice Disease Detection using Deep Learning

Aug 2025 – Dec 2025

Python, TensorFlow, EfficientNetV2, Computer Vision

GitHub | Live Demo

- Fine-tuned EfficientNetV2 on the Paddy Doctor dataset, reaching [X%] test accuracy across multiple disease classes – a [Y] point improvement over a baseline CNN.
- Shipped a Flask inference service serving predictions in under [Z]ms, used by [N]+ test farmers for real-time diagnosis.

NovaShield – AI Patient Health Monitoring System

Jul 2025 – Dec 2025

ESP32, TensorFlow Lite, Firebase, IoT

GitHub | Live Demo

- Built an edge-AI anomaly-detection pipeline on ESP32 (TensorFlow Lite) for heart rate, SpO₂, and temperature, flagging abnormal readings with [X%] precision.
- Reduced alert latency to under [Z] seconds via Firebase real-time sync, enabling remote monitoring for [N]+ simulated patients.

Smart Agriculture System using IoT and AI

Jan 2025 – Jul 2025

Python, ESP32, Machine Learning, Firebase

GitHub | Live Demo

- Built a real-time IoT sensor pipeline (soil moisture, temperature, humidity) across [N] field nodes feeding a crop-health prediction model.
- Optimized irrigation scheduling with the model's predictions, cutting estimated water usage by [X%] in simulation.

Additional Projects

- **Kinoklub** – Android app (Java/Kotlin, Firebase) for child-safe educational content with parental controls.
- **Medics** – Cross-platform appointment-booking app (Flutter, Dart, Firebase) with real-time sync.
- **Portfolio Website** – Responsive React site with optimized load performance.

Education

Brainware University – Master of Computer Application (MCA)

2024 – 2026

Barasat, West Bengal

Score: 81%

Vidyasagar University (Hijli College) – Bachelor of Computer Application (BCA)

2021 – 2024

Midnapore, West Bengal

Score: 80.02%

Achievements & Certifications

- 1st Prize – Paschim Medinipur District Level Science Fair (2023)
- 2nd Prize – IEEE Kharagpur Section Technical Competition, IIT Kharagpur (2022)
- [Add: LeetCode/Codeforces rating or problems solved, if any – highly weighted for FAANG screening]